# AFTER A V-ADMISSIBLE BLOWING UP A PROPER MORPHISM BECOMES AN ISOMORPHISM OVER A NEIGHBOURHOOD OF THE CLOSURE OF A CLOSED SUBSET WHERE IT WAS ALREADY ONE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0F3W (sources/stacks-flat/pinned-original-source.tex line 9608).

*Allowed standard prerequisites (editorial).*

- Separation of disjoint closed subsets of a spectral space by constructible closed subsets, and by an admissible blowing up. The spectral-space argument carried out at proof lines 9622-9629, and Same chapter, Lemma labelled lemma-separate-disjoint-locally-closed-by-blowing-up (source line 9554), invoked at proof lines 9630-9632. Supplies the constructible approximations T_c and T'_c with empty intersection and the blowing up making their closures disjoint.
- Zariski after blowup: an admissible blowing up whose strict transform is an isomorphism. Same chapter, Lemma labelled lemma-zariski-after-blowup (source line 9197), invoked at proof lines 9636-9638. Supplies the V_0-admissible blowing up Y'_0 -> Y_0 over which the strict transform of X_0 maps isomorphically.
- Extension of an admissible blowing up of an open to the whole scheme. Divisors, Lemma labelled divisors-lemma-extend-admissible-blowups (Stacks tag 080M), invoked at proof lines 9639-9641. Supplies the V-admissible blowup Y' -> Y restricting to the local solution on Y_0, which is what globalises the construction.

*Source reference.* Original lemma 0F3V: sources/stacks-flat/pinned-original-source.tex, source lines 9554–9554. Same-chapter pointer into the author environment lemma at source line 9554; invoked at proof line 9630 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 081S: sources/stacks-flat/pinned-original-source.tex, source lines 9197–9197. Same-chapter pointer into the author environment lemma at source line 9197; invoked at proof line 9638 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 080M: sources/stacks-chow/tags.txt, source lines 10187–10187. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9639 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *Let $f : X \to Y$ be a proper morphism of quasi-compact and quasi-separated schemes. Let $V \subset Y$ be a quasi-compact open and $U = f^{-1}(V)$. Let $T \subset V$ be a closed subset such that $f|_U : U \to V$ is an isomorphism over an open neighbourhood of $T$ in $V$. Then there exists a $V$-admissible blowing up $Y' \to Y$ such that the strict transform $f' : X' \to Y'$ of $f$ is an isomorphism over an open neighbourhood of the closure of $T$ in $Y'$.*

*Proof.* Let $T' \subset V$ be the complement of the maximal open over which $f|_U$ is an isomorphism. Then $T', T$ are closed in $V$ and $T \cap T' = \emptyset$. Since $V$ is a spectral topological space, we can find constructible closed subsets $T_c, T'_c$ with $T \subset T_c$, $T' \subset T'_c$ such that $T_c \cap T'_c = \emptyset$ (choose a quasi-compact open $W$ of $V$ containing $T'$ not meeting $T$ and set $T_c = V \setminus W$, then choose a quasi-compact open $W'$ of $V$ containing $T_c$ not meeting $T'$ and set $T'_c = V \setminus W'$). By Lemma 0F3V we may, after replacing $Y$ by a $V$-admissible blowing up, assume that $T_c$ and $T'_c$ have disjoint closures in $Y$. Set $Y_0 = Y \setminus \overline{T}'_c$, $V_0 = V \setminus T'_c$, $U_0 = U \times_V V_0$, and $X_0 = X \times_Y Y_0$. Since $U_0 \to V_0$ is an isomorphism, we can find a $V_0$-admissible blowing up $Y'_0 \to Y_0$ such that the strict transform $X'_0$ of $X_0$ maps isomorphically to $Y'_0$, see Lemma 081S. By Divisors, Lemma 080M there exists a $V$-admissible blow up $Y' \to Y$ whose restriction to $Y_0$ is $Y'_0 \to Y_0$. If $f' : X' \to Y'$ denotes the strict transform of $f$, then we see what we want is true because $f'$ restricts to an isomorphism over $Y'_0$. □